

CHAPTER

2

COMPLEX NUMBERS

2.1 Algebraic properties

Let (x, y) and (x', y') be two elements of $\mathbb{R}^2$. We define two operations on $\mathbb{R}^2$, by setting $(x, y) \times (x', y') = (xx' - yy', xy' + yx')$ et $(x, y) + (x', y') = (x + x', y' + y')$. This two composition operations define a new field, which is the complex commutative field denoted by $\mathbb{C}$. The additive neutral element is given by $0 = (0, 0)$ and the multiplicative neutral element is $(1, 0)$, the multiplicative inverse of $(x, y) \neq (0, 0)$ is $\left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right)$. By identifying $(x, 0) \in \mathbb{R}^2$ with $x \in \mathbb{R}$, and by setting $i = (0, 1)$.

$$\mathbb{C} = \left\{ z / z = x + iy \text{ with } x, y \in \mathbb{R} \text{ and } i^2 = -1 \right\}.$$

So we do calculus with complex numbers as what we do with real numbers taking into account that $i^2 = -1$.

Example 2.1. For all $n \in \mathbb{N}$, we have

$$1 + i + i^2 + \dots + i^n = \frac{1 - i^{n+1}}{1 - i} = \begin{cases} 1 & \text{if } n = 4k \\ 1 + i & \text{if } n = 4k + 1 \\ i & \text{if } n = 4k + 2 \\ 0 & \text{if } n = 4k + 3 \end{cases}$$

Definition 2.1. A complex number is any number of the form $z = a + ib$ where a and b are real numbers and i is the imaginary unit. The notations $a + ib$ and $a + bi$ are used interchangeably. The real number a in $z = a + ib$ is called the real part of $z = a + ib$; the real number b is called the imaginary part of $z = a + ib$. The real and imaginary parts of a complex number $z = a + ib$ are abbreviated $\text{Re}(z)$, and $\text{Im}(z)$, respectively.

Definition 2.2. Complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal, $z_1 = z_2$, if $a_1 = a_2$ and $b_1 = b_2$.

Conjugate and modulus of a complex number

Definition 2.3. Let $z = x + iy$ be a complex number, we define the conjugate of $\bar{z} = x - iy$ by $\bar{z} = x - iy$. The positive number $|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}$ denotes its modulus.

The modulus $|z|$ of a complex number z is also called the absolute value of z . We shall use both words modulus and absolute value throughout this text.

Example 2.2. If $z = 2 - 3i$, then we find the modulus of the number to be $|z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$. If $z = -9i$, then $|z| = |-9i| = \sqrt{(-9)^2} = 9$.

The following properties hold

- 1) $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$,
- 2) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$,
- 3) $\frac{\bar{z}_1}{z_2} = \frac{\bar{\bar{z}_1}}{\bar{z}_2}$.
- 4) $|z| \geq 0$ et $|z| = 0 \Leftrightarrow z = 0$,
- 5) $|z_1 z_2| = |z_1| |z_2|$
- 6) $||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$
- 7) $|\text{Re} z| \leq |z|$, $|\text{Im} z| \leq |z|$.
- 8) $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$.

2.2 Complex plane

A complex number $z = x + iy$ is uniquely determined by an ordered pair of real numbers (x, y) . The first and second entries of the ordered pairs correspond, in turn, with the real and imaginary parts of the complex number. For example, the ordered pair $(2, -3)$ corresponds to the complex number $z = 2 - 3i$. Conversely, $z = 2 - 3i$ determines the ordered pair $(2, -3)$. The numbers $7, i$, and $-5i$ are equivalent to $(7, 0), (0, 1), (0, -5)$, respectively. In this manner we are able to associate a complex number $z = x + iy$ with a point (x, y) in a coordinate plane.

Complex plane

Because of the correspondence between a complex number $z = x + iy$ and one and only one point (x, y) in a coordinate plane, we shall use the terms complex number and point interchangeably. The coordinate plane is called the complex plane or simply the z -plane. The horizontal or x -axis is called the real axis because each point on that axis represents a real number. The vertical or y -axis is called the imaginary axis because a point on that axis represents a pure imaginary number.

Vector

A complex number $z = x + iy$ can also be viewed as a two dimensional position vector, that is, a vector whose initial point is the origin and whose terminal point is the point (x, y) . This vector interpretation prompts us to define the length of the vector z as the distance $|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}$ from the origin to the point (x, y) . This length is the modulus.

2.3 Polar Form of Complex Numbers

Recall that a point P in the plane whose rectangular coordinates are (x, y) can also be described in terms of polar coordinates. The polar coordinate system, invented by Isaac Newton, consists of point O called the pole and the horizontal half-line emanating from the pole called the polar axis. If r is a directed distance from the pole to P and θ is an angle of inclination (in radians) measured from the polar axis to the line OP , then the point can be described by the ordered pair (r, θ) , called the polar coordinates of P .

Polar form

Suppose, that a polar coordinate system is superimposed on the complex plane with the polar axis coinciding with the positive x -axis and the pole O at the origin. Then x, y, r and θ are related by $x = r\cos\theta, y = r\sin\theta$. These equations enable us to express a nonzero

complex number $z = x + iy$ as $z = (r\cos\theta) + i(r\sin\theta)$. We say that $z = r(\cos\theta + i\sin\theta)$ is the polar form or polar representation of the complex number z . Again, the coordinate r can be interpreted as the distance from the origin to the point (x, y) . In other words, we shall adopt the convention that r is never negative so that we can take r to be the modulus of z , that is, $r = |z|$. The angle θ of inclination of the vector z , which will always be measured in radians from the positive real axis, is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an argument of z and is denoted by $\theta = \arg(z)$. An argument θ of a complex number must satisfy the equations $\cos\theta = \frac{x}{r}$ and $\sin\theta = \frac{y}{r}$. An argument of a complex number z is not unique since $\cos\theta$ and $\sin\theta$ are 2π -periodic. In practice we use $\tan\theta = \frac{y}{x}$ to find θ . However, because $\tan\theta$ is π -periodic, some care must be exercised in using the last equation. The following example illustrates how this is done.

Example 2.3. Express $-\sqrt{3} - i$ in polar form.

Solution

With $x = -\sqrt{3}$ and $y = -1$ we obtain $r = |z| = 2$. Now $\frac{y}{x} = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$, and so $\left(\tan\frac{1}{\sqrt{3}}\right)^{-1} = \pi/6$, which is an angle whose terminal side is in the first quadrant. But since the point $(-\sqrt{3}, -1)$ lies in the third quadrant, we take the solution of $\tan\frac{1}{\sqrt{3}}$ to be $\theta = \arg(z) = \frac{\pi}{6} + \pi = \frac{7\pi}{6}$. It follows that a polar form of the number is $z = 2(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6})$.

Principal Argument

The symbol $\arg(z)$ actually represents a set of values, but the argument θ of a complex number that lies in the interval $-\pi < \theta < \pi$ is called the principal value of $\arg(z)$ or the principal argument of z . The principal argument of z is unique and is represented by the symbol $\text{Arg}(z)$, that is, $-\pi < \text{Arg}(z) < \pi$. For example, if $z = i$, we have some values of $\arg(i)$ as $\frac{\pi}{2}, \frac{5\pi}{2}, \frac{-3\pi}{2}$, and so on, but $\text{Arg}(i) = \frac{\pi}{2}$. Similarly, we can verify that the principal argument of $-\sqrt{3} - i$ is $\text{Arg}(z) = \frac{\pi}{6} - \pi = \frac{-5\pi}{6}$. Using $\text{Arg}(z)$ we can express the complex number $-\sqrt{3} - i$ in the alternative polar form $z = 2\left(\cos\frac{-5\pi}{6} + i\sin\frac{-5\pi}{6}\right)$.

Moivre's formula

If $z \neq 0$ we have $z = r(\cos\theta + i\sin\theta)$ where θ is the principal argument. By definition, $-\pi \leq \text{Arg}(z) \leq \pi$. The polar form of a complex number is especially convenient when multiplying or dividing two complex numbers. We can verify that

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)).$$

Thus $|z_1 z_2| = |z_1| |z_2|$ and $\arg(z_1 z_2) = (\arg(z_1) + \arg(z_2)) \bmod 2\pi$,

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)).$$

continuing in this manner, we obtain a formula for the n -th power of z .

$$(r \cos \theta + i r \sin \theta)^n = r^n (\cos(n\theta) + i \sin(n\theta)).$$

Euler's formula

Let $z \neq 0$, and $z = r(\cos \varphi + i \sin \varphi)$, if $r = 1$, then $z = \cos \varphi + i \sin \varphi$. Put

$$e^{i\theta} = \cos \theta + i \sin \theta$$

which is called Euler's formula.

Example 2.4. $e^{2\pi i} = 1$, $e^{\pi i} = -1$, $e^{-\frac{\pi}{2}i} = -i$, $e^{\frac{\pi}{2}i} = i$.

Replacing θ by $(-\theta)$ in $e^{i\theta} = \cos \theta + i \sin \theta$, we obtain

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

From the last two formulas above, we deduce Euler's formulas:

$$\cos \varphi = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2}.$$

Properties

- 1) $e^{i\theta} \cdot e^{-i\theta'} = e^{i(\theta+\theta')}$
- 2) $\frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}$
- 3) $(e^{i\theta})^n = e^{in\theta}$, $n \in \mathbb{N}$.

Definition 2.4. Let $z \neq 0$. Then $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$ is called the exponential form of z .

The n th root of a complex number

All nonzero complex number $z = re^{i\theta}$ admits n roots n -ièmes w_k where, $w_k = \sqrt[n]{r} e^{i \frac{\theta+2\pi k}{n}}$ with $k \in \{0, 1, \dots, n-1\}$, $n > 1$.

Example 2.5. Solve the following equation $z^4 = 1 + i$.

The solutions are the fourth roots of $1 + i$, so we have $w_k = \sqrt[4]{2} e^{i \frac{\pi}{4} (\frac{\pi}{4} + 2\pi k)}$, $k = 0, 1, 2, 3$.